Yukawa screening derivation of the bond-valence rule

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The bond-valence model is a standard way to estimate bond strengths in crystals, but its exponential dependence on bond length has lacked a derivation from a specific physical interaction. We show that this form emerges as the leading-order limit of screened Coulomb electrostatics and that the fitted bond-valence softness can be interpreted in terms of an electronic screening length. This turns bond valence from an empirical fitting rule into a transferable descriptor of local screened charge response across coordination environments. The resulting theory predicts how the bond-valence parameters should vary with ionic charge and coordination number, and that prediction agrees with 150 fitted valences from 94 cation–oxygen species, including 68 in fourfold coordination and 82 in sixfold coordination, at an abundance-weighted coefficient of determination of 0.986. A comparison with first-principles charge densities shows that the bond-valence shell radius tracks the electronic screening cloud with coefficients of determination of 0.9998 for ten alkali and alkaline-earth oxides and 0.967 for 21 other binary oxides whose nearest-neighbor environments match the theory's assumptions. The widely used bond-valence model is thus the leading-order expression of screened electrostatics in ionic solids.

Pauling's observation that the strength of a bond reaching a coordinating anion approximates the cation charge divided by the coordination number [1] has been an organizing principle of crystal chemistry for nearly a century. Brown and Altermatt [2] cast this idea as the bond-valence model: the empirical bond strength between a cation i and anion j separated by R_{ij} is

$$S_{ij} = \exp\left(\frac{R_0 - R_{ij}}{B}\right), \tag{1}$$

where R_0 and B are parameters fitted to crystallographic data and the valence-sum rule $\sum_j S_{ij} = z_i$ (formal charge) is enforced. The model is used routinely for structure validation [3–5].

Several lines of work have sought a physical foundation for the bond-valence exponential. Preiser *et al.* [6] gave a clear Gauss's-law account, interpreting bond valence as localized electrostatic flux, and Brown developed this picture further within a broader electrostatic framework [3, 4]. However, in those treatments the exponential distance dependence is postulated rather than derived from a specific potential, leaving B as an empirical fit parameter with length units but no associated physical grounding. Adams related the softness B to Morse-potential curvature and motivated non-universal B values [7, 8], but does not connect B to a screening length. Modern first-principles work can compute screened local interactions from electronic structure [9, 10], but does not derive the bond-valence form itself. Here we show that a Yukawa expansion obtains the exponential as the leading term of a specific screened potential, ties B to a measurable screening length, predicts the B–R_0 correlation with no fitted constants, and connects the fitted parameters to electronic structure.

Screened-flux derivation. A cation with charge $Q_i = z_i e$ emits electrostatic flux Q_i/ϵ_0. In a Yukawa field with screening length λ, the signed flux amplitude captured by bond i–j is $\widetilde{\Phi}_{ij} \propto \Omega_{ij}\, e^{-R_{ij}/\lambda}(1 + R_{ij}/\lambda)$, where Ω_{ij} is the solid-angle fraction assigned to neighbor j. Bond valence is a nonnegative partition of the cation valence among its anion neighbors. Imposing the valence-sum rule $\sum_j S_{ij} = z_i$ and preservation of screened-flux magnitudes, $S_{ij}/S_{ik} = w_{ij}/w_{ik}$, uniquely gives

$$S_{ij} = z_i \frac{w_{ij}}{\sum_k w_{ik}}, \qquad w_{ij} := |\widetilde{\Phi}_{ij}|.$$

Choose R^* operationally as the Boltzmann-weighted mean bond length $m_i := \sum_j p_{ij} R_{ij}$ (defined fully in the partition-function section below). Expanding $\ln w_{ij}$ about R^* gives

$$\ln w_{ij} = C_i - \frac{R^*}{\lambda(\lambda + R^*)}\,(R_{ij} - R^*) + \delta_{ij}, \tag{2}$$

where C_i is independent of bond index j and $\delta_{ij} = O[(R_{ij} - R^*)^2/\{\lambda(\lambda + R^*)\}]$. Exponentiating the linear term and renormalizing therefore reproduces Eq. (1) at $R_{ij} = R^*$ with only quadratic log-weight error away from R^*. Define the leading-order Yukawa exponent denominator

$$B = \frac{\lambda(\lambda + R^*)}{R^*}. \tag{3}$$

This B is the first-order exponent denominator produced by the Taylor expansion of the Yukawa weight. On the screening branch it is positive and coincides with the conventional bond-valence parameter; on the anti-screening branch it becomes negative. The screening length λ is the primary physical variable, while B is its leading-order image in the bond-valence exponential. On the screening branch ($\lambda > 0$, $B > 0$), inverting Eq. (3) at the shell level gives the structure-specific screening

length $\lambda = (-R^* + \sqrt{R^{*2} + 4BR^*})/2$. At the species-level characteristic pair defined below, we evaluate the same positive-branch inversion on the characteristic reference shell, for which the operational choice $R^* = m_i$ gives $R^* = R_0^*$, so the tabulated species constant is $\lambda^* = (-R_0^* + \sqrt{R_0^{*2} + 4B^* R_0^*})/2$ for species with $B^* > 0$.

Parameter-free slope. In the uniform-bond limit, where $R_{ij} = \bar{R}$ for all j, the valence-sum rule gives $R_0 = \bar{R} + B\ln(z/n)$. Differentiating:

$$\beta := \frac{dB}{dR_0} = \frac{1}{\ln(z/n)}. \tag{4}$$

This is parameter-free: the slope of the B–R_0 correlation at fixed coordination number n is determined entirely by the charge-to-coordination ratio z/n, with a pole at $z = n$ marking the screening/anti-screening crossover. Figure 1 shows that Eq. (4) reproduces the fitted slopes for 94 cation–oxygen species at $n = 4$ and $n = 6$ (68 at $n = 4$, 82 at $n = 6$; 150 slopes in total; weighted $R^2 = 0.986$, mean absolute error 0.15; weighting scheme defined in the Supplemental Material [11]).

Screening collapse at $z = n$. The pole has direct physical content. At $z = n$ every bond carries exactly one valence unit ($S_{ij} = 1$), so $\exp[(R_0 - R_{ij})/B] = 1$ for all j. For a symmetric shell with $R_{ij} = \bar{R}$ this is trivially satisfied at any B with $R_0 = \bar{R}$—the $z = n$ limit is degenerate. For a real coordination shell with unequal bond lengths, however, the only solution is $B \to \infty$, so that every finite $(R_0 - R_{ij})/B \to 0$. On the screening branch, Eq. (3) gives $\lambda \approx \sqrt{BR^*} \to \infty$: the screening length diverges. On the $z > n$ side of the pole, B changes sign ($B < 0$). Its magnitude still diverges, so the branch change is carried directly by $\operatorname{sgn} B$ or, equivalently, by the sign reversal of $\beta = 1/\ln(z/n)$. Approaching the pole from either side ($z = n \mp \varepsilon$, $\varepsilon \to 0$), the divergence rate is governed by the species characteristic intercept R_0^* (the n-resolved-line convergence point introduced below):

$$|B| \sim \frac{n\,|R_0 - R_0^*|}{\varepsilon}, \qquad |\lambda| \sim \sqrt{\frac{R^*\,n\,|R_0 - R_0^*|}{\varepsilon}}.$$

The screening length diverges as $\varepsilon^{-1/2}$. In conventional screening theory, metals have finite Thomas–Fermi screening set by a nonzero density of states at the Fermi level, whereas gapped or correlated insulators suppress low-energy charge response and therefore screen more weakly [12–15]. Our coarse-grained λ is not identical to those microscopic lengths, but its pole at $z = n$ is consistent with the same weak-screening limit: the bond-valence partition becomes insensitive to finite bond-length differences and the fitted B–R_0 parameters cease to resolve the shell. Species that sit at or near $z = n$—Si^{4+} in tetrahedral coordination (SiO_2, band gap 9 eV), B^{3+} in trigonal coordination (B_2O_3, 6 eV), and C^{4+} at $n = 4$—lie in the wide-gap covalent limit where such weak screening is physically plausible.

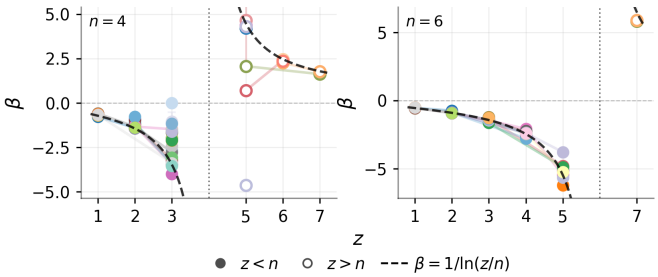

FIG. 1. Fitted slope β vs. formal charge z for 94 cation–oxygen species at $n = 4$ (68 cations, left) and $n = 6$ (82 cations, right); 56 cations contribute at both coordinations, giving 150 fitted slopes in total. Dashed curves: $\beta = 1/\ln(z/n)$. The pole at $z = n$ separates the screening ($z < n$) and anti-screening ($z > n$) branches; species at $z = n$ are omitted because the B–R_0 fit is degenerate at the pole (see Supplemental Material [11]).

Partition thermodynamics. Because R_0/B cancels in $S_{ij}/\sum_k S_{ik}$, the normalized bond weights $p_{ij} = S_{ij}/\sum_k S_{ik} = e^{-R_{ij}/B}/Z_i$ define a canonical partition function with $B > 0$ as the temperature-like scale on the screening branch. The Boltzmann mean $m_i := \sum_j p_{ij} R_{ij}$ and the Shannon entropy $H_i := -\sum_j p_{ij} \ln p_{ij}$ yield the exact identity

$$R_0 = m_i + B(\ln q_i - H_i),$$

where $q_i := \sum_j S_{ij}$. Here H_i is a purely geometric quantity: it measures how uniformly the bond-valence weight is distributed across the coordination shell, reaching $\ln n$ for a symmetric shell and falling below it as distortion concentrates weight on the shorter bonds. Thus R_0 is not the mean bond length but an entropy-corrected weighted mean, and the operational choice $R^* = m_i$ in Eq. (3) is made explicit.

Empirical validation. Bond-valence parameters (R_0, B) were determined for 103 cation–oxygen species using first-principles relaxed crystal structures from the Materials Project [16, 17]. For each structure, bond strengths were obtained by solving the valence-sum and loop constraints on the cation–oxygen bond graph [18], and (R_0, B) were fitted by linear least squares; the few rank-deficient bond graphs and numerically ill-conditioned fits were treated by the procedure described in the Supplemental Material [11]. The resulting set spans s-, d-, f-, and p-block cations; Fig. 1 shows the 94 species (150 fitted slopes) for which the prediction is finite at $n = 4$ and/or $n = 6$.

Within each (z, n) group, the slope $\beta^{(n)}$ confirms Eq. (4) across all families. The n-resolved lines converge to cation-specific characteristic pairs (R_0^*, B^*), defined as the intersection that minimizes the weighted variance of the coordination-resolved B–R_0 lines, with each line weighted by the final fit population of that coordination shell (Supplemental Material [11]). Of the 103 species,

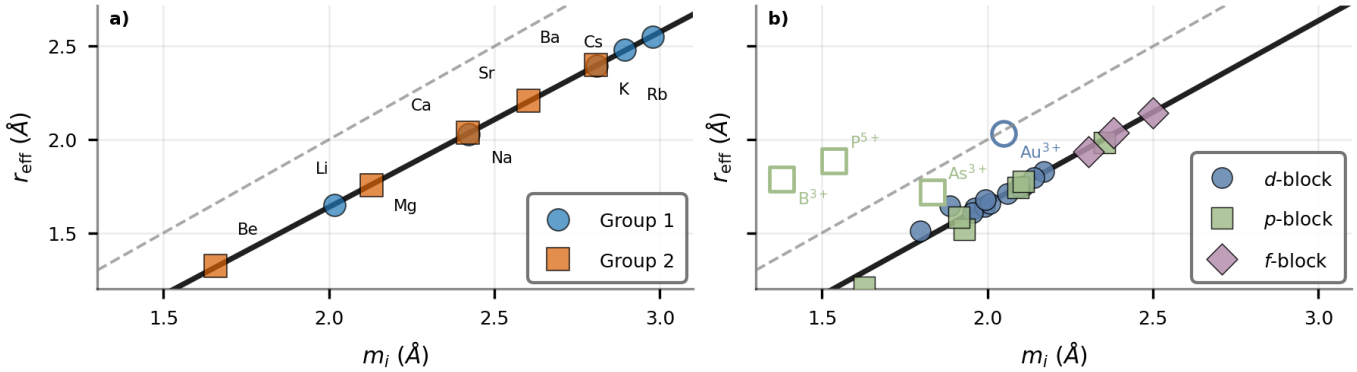

FIG. 2. First-principles screening centroid $r_{\rm eff}$ vs. Boltzmann shell centroid m_i. (a) Ten s-block ground-state binary oxides. Solid line: shared fit; dashed line: $r_{\rm eff} = m_i$. (b) Twenty-five non-s-block ground-state binary oxides, color-coded by block. Open symbols mark the four directional-bonding outliers excluded from the regression (B_2O_3, P_2O_5, As_2O_3, Au_2O_3).

101 have $B^* > 0$ (screening branch); for these the screening length $\lambda^* = (-R_0^* + \sqrt{R_0^{*2} + 4B^* R_0^*})/2$ is a tabulated constant (Supplemental Material [11]). The remaining two species (Mo^{3+}, Sb^{3+}) have $B^* < 0$, fall on the anti-screening branch, and are retained only as signed diagnostics. Both sit outside the first-order Yukawa regime: Sb^{3+} has a stereoactive $5s^2$ lone pair driving strong directional distortion, and Mo^{3+} is constrained by only two coordination shells (Supplemental Material [11]), so its (R_0^*, B^*) is provisional. The screening-branch B^* values are non-universal (mean 0.38 ± 0.15 Å), in qualitative agreement with Adams's softBV parameterization [7], while λ^* provides a single-parameter species descriptor.

First-principles confirmation. The connection to electronic structure is tested by comparing the bond-valence Boltzmann shell centroid $m_i = \sum_j p_{ij} R_{ij}$ to a first-principles screening centroid $r_{\rm eff}$ extracted from charge densities in the Materials Project [16, 17]. For each M–O bond with $R_{ij} \leq 3.5$ Å, the total valence-electron density $\rho(r)$ is sampled at 200 uniformly spaced points along the nearest-image straight M–O segment. An interior density-minimum proxy on $0.15R_{ij} < r < 0.85R_{ij}$ defines the start of the oxygen-side window; $r_{\rm eff}$ is then the excess-density-weighted centroid on $r_{\rm BCP} \leq r \leq 0.93R_{ij}$ with weights $\max[\rho(r) - \rho_{\rm BCP}, 0]$, averaged over the valid first-shell bonds of that oxide (see Supplemental Material for the exact discrete construction and fallback rule). For ten s-block binary oxides (Li_2O through BaO; Fig. 2a), these collapse onto a single line:

$$r_{\rm eff} = 0.94\, m_i - 0.24 \text{ Å} \qquad (R^2 = 0.9998).$$

This is not a trivial monotone size trend: the jackknife mean absolute error of the m_i regression is 0.006 Å, versus 0.078 Å for a linear fit against R_0 and 0.094 Å against the Shannon cation radius [19] (see Supplemental Material for the corresponding control plot). The near-unit slope therefore shows that the bond-valence partition function and the first-principles screening cloud track the same ionic shell radius. The same centroid correspondence extends beyond the s block. Figure 2b shows the 25 non-s-block binary oxides for which the charge density yields a well-defined oxide-level screening centroid (listed in the Supplemental Material [11]). The Yukawa derivation assumes approximately isotropic first-shell screening, so the regression is restricted to the 21 oxides whose cation environments satisfy that condition, excluding four whose bonding is dominated by directional, non-isotropic motifs: B_2O_3, P_2O_5, As_2O_3, and Au_2O_3; the Supplemental Material gives the fixed inclusion rule and the exact oxide-level centroid construction used to define $r_{\rm eff}$. The 21 included oxides follow

$$r_{\rm eff} = 0.98\, m_i - 0.30 \text{ Å} \qquad (R^2 = 0.967).$$

The centroid correspondence therefore extends across the p-, d-, and f-block with $R^2 = 0.967$, and its breakdown is confined to strongly anisotropic directional-bonding environments outside the isotropic Yukawa regime.

Discussion. The main result is that the bond-valence exponential—the workhorse of crystal-chemistry validation for four decades—is the leading-order term of Yukawa screened electrostatics. The fitted exponent denominator B is not a free parameter but the leading-order Yukawa denominator set by the screening length and shell geometry, with its sign carrying the screening/anti-screening branch information; the parameter correlation $\beta = 1/\ln(z/n)$ is parameter-free and exact in the uniform-bond limit; and the partition-function structure gives an entropy-corrected identity for R_0 and a free energy $F = m_i - BH_i = R_0 - B \ln q_i$ in length units.

The first-principles centroid comparison confirms that this is not merely a mathematical analogy: the electronic screening cloud and the bond-valence weights describe the same shell geometry. The small deviation from the identity line is explained by the oxygen-side valence centroid shift measured directly from the same charge-density paths, while the null-model controls show that

the collapse is substantially sharper than a generic monotone size trend.

The derivation rests on two approximations: (i) the first-order Taylor expansion, which breaks down for highly distorted shells (d^4/d^9 Jahn–Teller octahedra), and (ii) the assumption that solid-angle fractions Ω_{ij} are approximately uniform across the coordination shell. Solid-angle non-uniformity contributes an additive $\ln \Omega_{ij}$ term to Eq. (2) that is absorbed into C_i for near-regular polyhedra but partially decorrelates the bond weights from λ alone when the shell departs from local symmetry, as in stereoactive lone-pair and strongly anisotropic environments [4, 20]. The empirical signature of this breakdown is visible in Fig. 2b: the four oxides whose ground-state structures fail the isotropic-shell assumption (B_2O_3, P_2O_5, As_2O_3, Au_2O_3) are precisely those that depart from the centroid line, identifying ground-state directional bonding as the empirical boundary of the first-order Yukawa regime. Extending the first-principles centroid comparison to the full set of transition-metal and f-block oxides beyond the subset treated here will require going beyond the Thomas–Fermi approximation to include orbital directionality and correlation effects.

Bond valence is therefore recast as a coarse-grained observable of local screened charge response rather than a purely empirical structural heuristic. Concretely, the species-specific screening length λ^* is a compact tabulated descriptor that organizes the shell thermodynamics and, through the identities above, provides a calibrated starting point for estimating bond weights, entropy, effective coordination number, and R_0 in new coordination environments without additional adjustable parameters. Establishing the full predictive scope of that program beyond the families validated here remains future work. This connects Pauling's 1929 valence rules [1] to Yukawa's 1935 screened potential [21] through a quantitative bridge that has been missing for nearly a century.

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